Math Inquirers Primary 4: Scheme of Work

Holiday program for ages 9 to 11 (builds on Primary 4 maths) • 10 lessons of 90 minutes • 2 lessons a week for 5 weeks • live online, up to 8 children

Approach. Enrichment, not extra drill: non-routine puzzles on topics chosen to complement the school syllabus. Lessons 1 to 9 follow the same routine, in minutes of the 90: Opening (1.5), sneak-peek puzzles (5 to 5.5), warm-up (5 to 5.5), explore (in pairs) (11.5 to 14.5), mini-lesson (5 to 8.5), brain break (3.5 to 4), worksheet (42.5 to 48), talk time (3.5 to 4), exit ticket (2.5 to 4), and looking back (3.5). Lesson 10 has its own plan. The time plan under each lesson is the one in the tutor guide.
Habits. Notice it. Test it. Explain why.

Lesson	Title and school topic	Objective	Skills and heuristics practised	Exit ticket
1 Week 1A	**Factor Hunters** Factors and multiples	Find all the factor pairs of a number by working in order, find common factors and common multiples to solve sharing and timetable puzzles, discover which numbers have an odd number of factors, and solve multiple-based logic riddles.	Making a list in order, looking for a pattern, using clues one at a time, explaining why	Write all the factors of 28. Then find the smallest number that is a multiple of both 6 and 9.
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5.5** · Warm-up **5.5** · Explore (in pairs) **12** · Mini-lesson **8** · Brain break **3.5** · Worksheet **43.5** · Talk time **3.5** · Exit ticket **3.5** · Looking back **3.5**			
2 Week 1B	**Big Number Puzzles** Numbers to 100 000	Round numbers to the nearest 10, 100 and 1000 using a number line, work backwards from a rounded number to find the smallest and largest possible numbers, use rounding to estimate and check, and solve 5-digit number riddles, digit-card puzzles and number patterns.	Using a number line, working backwards, making a list in order, using the clues one at a time, looking for a pattern	Round 74 652 to the nearest thousand. Then write the smallest whole number that rounds to 74 000 to the nearest thousand.
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **13** · Mini-lesson **5** · Brain break **3.5** · Worksheet **47** · Talk time **3.5** · Exit ticket **3** · Looking back **3.5**			
3 Week 2A	**Operation Detectives** Four operations	Use the order of operations (brackets first, then multiplication and division from left to right, then addition and subtraction from left to right), find missing signs and brackets to hit a target, work backwards through a chain of operations, and solve "times as many" problems with a bar model.	Order of operations, estimating then testing, guess and check, working backwards, drawing a bar model, explaining why a number trick works	Work out $50 - 6 \times (3 + 4)$.
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **12** · Mini-lesson **5** · Brain break **4** · Worksheet **48** · Talk time **3.5** · Exit ticket **2.5** · Looking back **3.5**			
4 Week 2B	**Fraction Frontier** Fractions	Change between improper fractions and mixed numbers using fraction bars and number lines, find a fraction of a set (and the whole set from a part) with a bar model, and compare fractions with different denominators by making the denominators the same, comparing with one half, or comparing pieces of the same numerator.	Drawing a fraction model, using a number line, drawing a bar model, making a list in order, comparing with a benchmark, explaining a comparison	Write 23/5 as a mixed number. Then find 3/5 of 25.
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **12.5** · Mini-lesson **8** · Brain break **3.5** · Worksheet **44** · Talk time **4** · Exit ticket **3** · Looking back **3.5**			
5 Week 3A	**Decimal Detectives** Decimals	Read decimals to 3 places on number lines, compare and order them by place value, round to the nearest whole number, 1 or 2 decimal places and work backwards from a rounded number (smallest and largest), and solve money and mass puzzles with bar models.	Place value to thousandths, zooming in on a number line, ordering, rounding and rounding backwards, systematic listing, bar models, checking every clue	Write 2.8, 2.08 and 2.088 in order, smallest first. Then round 2.088 to 1 decimal place.
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **13.5** · Mini-lesson **8** · Brain break **3.5** · Worksheet **43** · Talk time **3.5** · Exit ticket **3.5** · Looking back **3.5**			

Lesson	Title and school topic	Objective	Skills and heuristics practised	Exit ticket
6 Week 3B	**Angle Turns** Angles	Estimate and measure angles with a protractor, find every angle in a fan of rays, work out angles between clock hands (30° for each number) and turns of the minute hand (6° a minute), turn on the 8-point compass (45° steps) and work backwards from a turn, and find unknown angles in squares and rectangles.	Estimating with 90° and 180°, measuring, systematic counting, adding and subtracting angles, working backwards, explaining with turns	Face N. Turn 135° clockwise. Which way do you face? What is the angle between the clock hands at 5 o'clock?
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **11.5** · Mini-lesson **7.5** · Brain break **3.5** · Worksheet **46** · Talk time **3.5** · Exit ticket **3** · Looking back **3.5**			
7 Week 4A	**Area Architects II** Area and Perimeter	Find the area of a composite figure by splitting it into rectangles or by taking away a missing piece, find its perimeter by pushing the steps out to a rectangle, find every rectangle with a given area and compare their perimeters, and explain what changes when a shape is cut and rearranged.	Splitting and taking away, making a list in order, looking for a pattern, explaining why (area stays the same, perimeter can change)	A 7 cm by 5 cm rectangle has a 2 cm by 2 cm square cut from one corner. Find the area and the perimeter of the shape that is left.
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **12** · Mini-lesson **5.5** · Brain break **3.5** · Worksheet **46** · Talk time **4** · Exit ticket **4** · Looking back **3.5**			
8 Week 4B	**Symmetry and Nets** Symmetry and Nets	Decide whether a shape made of 6 squares folds into a cube by folding it in your head, find which faces of a net are opposite (and use it for dice), complete a figure so that a given line is a line of symmetry, find all the lines of symmetry of a shape, and shade squares to make a pattern symmetric.	Visualising (folding in your head), making a list in order, testing every case, explaining why	This dice net shows 1, 3 and 2. Which number goes in the square marked "?"
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **14** · Mini-lesson **8.5** · Brain break **3.5** · Worksheet **42.5** · Talk time **4** · Exit ticket **2.5** · Looking back **3.5**			
9 Week 5A	**Data Detectives** Tables and line graphs	Read values and trends from line graphs (using the scale), find the biggest increase or decrease, find missing values in tables and graphs from totals and patterns, compare two lines on one graph, and explain why a graph with a different scale can mislead.	Reading scales, comparing changes, working from totals, reasoning about how a graph is drawn	Visitors: Mon 35, Tue 50, Wed 20, Thu 45. Between which two days was the biggest change? Was it an increase or a decrease?
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **14.5** · Mini-lesson **6.5** · Brain break **3.5** · Worksheet **44** · Talk time **3.5** · Exit ticket **3** · Looking back **3.5**			
10 Week 5B	**Puzzle Day** Mixed heuristics	Choose a way into a non-routine problem (guess and check with a table, suppose they are all the same, work backwards, a logic grid, draw it) and use it on competition-style puzzles, checking every answer against every fact.	Choosing a strategy, guess and check, supposition, working backwards, logical deduction, checking and explaining	A farm has ducks and goats: 12 heads and 34 legs. How many goats? Show your way in.
	Time plan (minutes): Opening **1.5** · Sneak-peek puzzles **5** · Warm-up **5** · Explore (in pairs) **13** · Mini-lesson **6.5** · Brain break **3.5** · Worksheet **41.5** · Talk time **3.5** · Exit ticket **4.5** · Looking back **3.5** · Certificates and goodbye **2.5**			

Assessment and progress

Each lesson has a 2-page Start here worksheet, then Think harder, Go further and an Investigation with a Look-back (printed for every child, who may choose any of them at any time; nobody is placed in a group), an exit ticket and an Inquiry Journal stamp. Tutors record S (secure), D (developing) or H (needs help) for each child on the progress tracker in the tutor guide. Parents receive a short note each week and the Weekly Dose of Thinking Math practice pack.

Materials

Printed student workbook with cut-outs, pencil, scissors and paper; a laptop or tablet with a camera. Tutors use the slides for each lesson and the tutor guide scripts.